

Priyansh Mahajan

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EDUCATION

- **Imperial College London** London, UK
Computing – Integrated Master’s (MEng); First-Class Honours (80% overall) 2021 – 2025
 - Awarded **Dean’s List Prize** in 2nd and 3rd years, placing within the top 10% of the computing cohort.
 - Awarded **ARM Best Overall C Project Prize** (2021/22) for low-level system design with a score of 93%.
- **Tiffin Grammar School** 2014 – 2021
 - **A-Levels:** A*s in Mathematics, Further Mathematics, Physics, and Economics.
 - **Mathematics Olympiads:** Awarded UKMT **Book Prize** (Top 50 UK mathematicians nationally), BMO Round 1 **Distinction**, and consecutive annual UKMT Olympiad Gold Certificates.

EXPERIENCE

- **Palantir Technologies** London, UK
Software Engineer (Forward Deployed & Core Product) August 2025 – Present
 - Engineered core frontend and full-stack features for a new product showcased live on the **Main Stage at Palantir AIP DevCon**, delivering technical briefings directly to C-Suite executives.
 - Led technical scoping and problem decomposition for Startup Fellowship partners, managing day-to-day technical operations and preparing partner startups for public DevCon presentations.
 - Productionized an enterprise document processing engine on Palantir Foundry, implementing asynchronous chunking and distributed batching pipelines to scale throughput from **1.5 to 300 pages/min**.
- **Elliott Advisors** London, UK
Software Engineer (Intern & Part-Time) April 2024 – March 2025
 - Integrated distributed request tracing via Datadog APM into the core **Order Management System (OMS)** in C#, profiling HTTP and SQL request latency metrics to pinpoint critical execution bottlenecks.
 - Led development of a greenfield C# system replacing legacy compensation infrastructure, collaborating directly with upper management on system architecture and deployment.
- **Ministry of Defence** London, UK
Software Engineer Intern June 2023 – Sept 2023
 - Engineered a full-stack React and AWS application leveraging Amazon Rekognition to extract video objects of interest, architecting parallel Python/AWS Lambda pipelines to cut end-to-end processing latency by **70%**.

PROJECTS

- **LLVM Memory Data Prefetching Loop Cloning**
Master’s Thesis in Compiler-Microarchitecture Co-Design & Simulation
 - Developed custom **LLVM IR compiler passes** in C++ to perform static analysis and automated loop cloning for Memory Data Prefetching (MDP).
 - Evaluated microarchitectural cache miss reductions and memory bandwidth efficiency on out-of-order CPU models using the **gem5 architectural simulator** and SPEC CPU benchmarks.
- **C++ Path Tracer**
 - Architected a zero-allocation CPU path tracing engine in C++23 featuring a custom Surface Area Heuristic (SAH) BVH with division-free AABB ray-box intersections (overall 46.8x speedup).
 - Implemented tile-based multithreading via **Intel TBB** (8.2x speedup), eliminating `std::shared_ptr` atomic ref-counting and virtual function calls via `std::variant` / `std::visit` dispatch.
- **Hash Join Engine**
Snowflake Data Processing Systems Competition Winner (Top 3 Finish)
 - Designed a high-throughput multithreaded hash join algorithm in C++, employing a Decomposed Storage Model (DSM / columnar) for CPU L1/L2 cache locality and optimized multithreaded hash table probing.
- **WACC Language Compiler**
 - Built a full compiler pipeline in Haskell translating WACC source code to ARM assembly, implementing AST generation, type checking, register allocation, and stack frame management.

TECHNICAL SKILLS

- **Programming Languages:** C++, C, Rust, Haskell, C#, Python, TypeScript, SQL, Assembly (x86/ARM).
- **Tools & Infrastructure:** LLVM, Docker, AWS, React, PostgreSQL, Linux, Git.